

LINDSAY CHENG

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EDUCATION

Stanford University

M.S. Computational & Mathematical Engineering

Septembmer 2017 - June 2019

GPA: 3.61

University of California, Los Angeles

B.S. Applied Mathematics and B.S. Neuroscience.

July 2013 - June 2017

GPA: 3.71

Honors: Cum Laude, Deans List, ALD | PES Honor Society, Golden Key International Honor Society

RELEVANT EXPERIENCE

Google Deepmind

Senior Software Engineer

September 2023 - Present

Mountain View, CA

- Designed and implemented the orchestration infrastructure for long-running agents and tool-calling, optimizing the interaction between LLM reasoning traces and external API execution.
- Improved model quality and inference performance for Gemini-based UI control products, focusing on intent-to-action mapping and system-level prompt optimization.
- Drove quality engineering for universal agentic planner experiences on agency framework, establishing evaluation benchmarks for multi-step reasoning and complex task execution.
- Partnered with Research teams to translate advanced model training techniques into production-ready, low-latency models for functional UI automation.
- Designed and implemented tree-search algorithms for browsing agents to improve navigation reliability and long-horizon goal completion.

Snowflake Inc (Acquired Neeva)

Software Engineer - Snowflake AI

May 2023 - August 2023

San Mateo, CA

- Drove architectural decisions for large-scale embedding generation and dense retrieval systems, optimizing vector-based data programmability for enterprise-scale AI.
- Managed concurrent model training and fine-tuning jobs for the Snowflake Core Copilot, triaging issues and focusing on specialized optimizations for SQL generation and code-centric tasks.
- Aligned cross-functional data pipelines and training criteria with real-world customer usage to improve LLM utility and instruction-following in production.

Uphonest Capital

Scout

April 2023 - Present

Santa Clara, CA

- Assist Uphonest Capital in discovering hidden, exceptional startup, mainly in AI, search, Saas, health and fan-art.

Neeva

Software Engineer, Machine Learning and Search Ranking

December 2019 - May 2023

Mountain View, CA

- Fine-tuned and deployed LLMs in production to enhance search result ranking, significantly increasing click-through rates and user satisfaction.
- Pioneered personalization efforts for the Neeva Home and Search Results pages, leveraging user signals to prioritize relevant content.
- Enhanced the quality and efficiency of the crawling service, and indexing within the core system, optimizing search performance.
- Significantly boosted local search quality through targeted improvements in query understanding and results ranking.
- Developed an automated content generator based on user interests, utilizing third-party API integrations to drive platform engagement.
- Architected and built the dense retrieval system within the core search engine infrastructure, moving beyond traditional keyword matching.
- Mentored early-career engineers and interns, establishing best practices for the ML engineering team during rapid growth phases.

Acrobat Genomics

Associate Computational Biologist (part-time)

March 2023 - July 2023

San Carlos, CA

- Derived underlying mathematical parameters and managed configurations to build an AI generative model for a protein engineering platform.

Stanford University School of Medicine, Neurobiology Department

Graduate Research Assistant

April 2019 - December 2019

Palo Alto, CA

- Represented neural mechanism using deep learning model given electrical step inputs
- Conducted the experiment to reveal the intertwined neural signal information using Recurrent Neural Network model
- Extended the system into a more in-depth dynamics which represented the human neural activities during prediction of environmental changes

Stanford University, Psychology Department
Graduate Research Assistant

January 2019 - June 2019
Palo Alto, CA

- Studied mathematical cognition and the development of human abilities to obtain mathematical perceptions
- Designed a deep neural network model to study how human learn magnitude comparison and simple addition for one-digit numeral
- Analyzed and interpreted how approximate arithmetic training may affect preschoolers performances in simple math problems

SAP Labs, LLC

iXp Intern - Applied Data Science and Machine Learning (NLP), Recast Conversational AI

June 2018 - August 2018
Palo Alto, CA

- Helped create a platform for collaborative chatbots satisfying various needs from customers
- Tested 50+ different combinations of encodings and clustering algorithms on data with ground truth
- Applied the best algorithm to cluster the data without ground truth and output the results based on how dense the clusters are, and collaborated with the front-end team so that users can easily visualize the suggested sentences based on intents when creating chatbots
- Incorporated more embeddings and similarity computation methods into the existing algorithms

UCLA Math Department

Programming in Computing Reader

September 2016 - June 2017
Los Angeles, CA

- Helped instructors create the grading rubrics
- Graded homework in Introduction to Programming (C++) course and provided feedback to students about their algorithm

ADDITIONAL EXPERIENCES

Deloitte Enterprise Consulting (Shanghai) Co., Ltd. Shenzhen Branch

Enterprise Risk Service, Financial Service Industry Intern

Summer 2016
Shenzhen, China

- Reviewed the manual of constructing online risk-management systems for companies, and edited the manual according to requests from customers
- Collected risk-management information from annual reports of various securities companies, and organized information for supervisors reference
- Discussed and consulted with group members best modeling strategies for every particular case

UCLA David Geffen School of Medicine, Neurology Department

Undergraduate Research Assistant

January 2016 - June 2017
Los Angeles, CA

- Studied the pathology of Parkinsons Disease by conducting behavioral experiments on zebrafish with a small team
- Monitored the movements of zebrafish using Viewpoint software to focus on transgenic genes pathological in Parkinsons Disease
- Analyzed the behavior data, including over 10,000 speed data, which reflects instantaneously the motion of the fish above a certain threshold
- Imaged the neuron distributions inside the fish bodies and counted any significant loss of neurons

SKILLS

Programming Languages

Python, Golang, C++, Julia

Tools

Kubernetes, Databricks, Spark, Tensorflow, Pytorch, Airflow, Grafana

PUBLICATIONS

Sizhu Cheng, Arianna Yuan. *Understanding the Learning Effect of Approximate Arithmetic Training: What is Actually Being Learned?*

Paper accepted for the 17th Annual Meeting of the International Conference on Cognitive Modeling, Montreal, Canada

Sataree Khuansuwan, Lisa M. Barnhill, **Sizhu Cheng**, and Jeff M. Bronstein. *A novel transgenic zebrafish line allows for in vivo quantification of autophagic activity in neurons*, Autophagy 2019

SERVICE ACTIVITIES

Stanford Women in Math Mentoring, Stanford University

Mentor

Fall 2017 - Present
Palo Alto, CA

- Provides advises to women undergraduates who consider graduate school in mathematical fields
- Assists women undergraduates in arranging their undergraduate studies and facilitates them to find their career goals

Global Medical Training, UCLA

Medical Donation Committee

Spring 2015 - Spring 2017

Los Angeles, CA

- Contacted past donors, including local hospitals, health centers, and other organizations for medical supplies
- Worked with fundraising team to raise funds to purchase the necessary medications and medical equipment
- Searched and applied for grants to raise funds for the committee and wrote applications when necessary